# Anthi Voulgari Revof, M.Sc.


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## About me

- I am currently a PhD student at University of South-Eastern Norway, studying Quantum Reference Frames. I completed the joint Master's program in High Energy Physics at IP Paris and ETH Zurich, graduating with highest honors.

## Education

09/2024 – 08/2025
*Zurich, Switzerland*

**Master 2 High Energy Physics at *ETH Zurich* .**

- Coursework includes: *QCD and Scattering Amplitudes, Effective Field Theories for Particle Physics, String Theory, Introduction to Integrability*
- Thesis: *"Gravitational Memory effect in de-Sitter spacetime"* supervised by Dr. Shubhanshu Tiwari and Prof. Philippe Jetzer (UZH).
- ECTS: 60

09/2023 – 09/2024
*Palaiseau, France*

**Master 1 High Energy Physics at *Institut Polytechnique de Paris* .**

- Coursework includes: *Perturbative Quantum Field Theory I & II, Particle Physics I & II, General Relativity, Symmetry Groups in Subatomic Physics, Astrophysics*
- Internship: *"Bayesian Machine Learning Algorithm for event reconstruction at HK"* supervised by Dr. Lorenzo Perisse.
- ECTS: 60

09/2019 – 07/2023
*Patras, Greece*

**Bachelor of Physics at the *University of Patras, Department of Physics* .**

- Specialization: Theoretical, Computational Physics and Astrophysics
- Thesis: *"Symmetries in Quantum Field Theory and General Relativity"*, supervised by Prof. Andreas Terzis.
- ECTS: 240
- Final Grade: 9.35/10

## Projects and Training

02/2025 – 09/2025
*Zurich, Switzerland*

**Master's Thesis *"Gravitational Memory effect in de Sitter spacetime and detection possibilities with LISA"*.**

- Computed displacement and spin memory observables in de Sitter spacetime, including explicit $\Lambda$-corrections and multipolar expansions.
- Analyzed waveform modeling of compact binary black hole mergers.

09/2024 – 11/2024
*Zurich, Switzerland*

**Proseminar at ETH Zurich, Institute for Theoretical Physics (ITP), held by Prof. Charalampos Anastasiou.**

- Focused on linear and non-linear sigma models as an EFT framework for understanding Higgs physics in Electroweak Symmetry Breaking.
- Examined the utility of non-linear field representations for decoupling heavy modes while preserving chiral symmetry, leading to Chiral Perturbation Theory.

04/2024 – 08/2024
*Tokyo, Japan*

**Internship at ILANCE, University of Tokyo, Institute of Cosmic Ray Research.**

- Employed a **Bayesian Machine Learning Algorithm** to reconstruct the energy of electron events in **Hyper Kamiokande**.
- Implemented a **3-layer Bayesian Neural Network (BNN)**, trained with approximately **300,000 events**.
- Achieved an uncertainty of around 3 MeV and investigated whether the predicted uncertainty accurately reflected the true event uncertainty.

01/2024 – 04/2024

## Projects and Training (continued)

*Paris, France*
09/2022 – 07/2023

- **Research Project at IP Paris, Centre de Physique Theorique (CPHT), supervised by Prof. Marios Petropoulos and Prof. Guillaume Bossard**.
  - Explored the **ADM (Arnowitt-Deser-Misner)** formulation of General Relativity and its application to the **asymptotic structure** of spacetime at spatial infinity.
  - Investigated the symmetries of spacetime, including **BMS symmetry group**, and addressed the smoothness conditions required at null infinity.

*Patras, Greece*

- **Bachelor's Thesis *"Symmetries in Quantum Field Theory and General Relativity"***.
  - Investigated symmetry-breaking scenarios in Quantum Field Theory and General Relativity, with emphasis on the question whether GR can be seen as a gauge theory.
  - Analyzed invariance under infinitesimal diffeomorphisms in linearized GR and demonstrated the existence of a coordinate system where the perturbation metric satisfies the Lorentz gauge.

## Technical Skills

Languages
- Strong reading, writing, and speaking competencies (C2) in **English** and **French**. My native language is **Greek**.

Coding
- Python, Matlab, Wolfram Mathematica, C++, LaTeX

## Honors and Awards

09/2024 – ongoing
- **Scholarship support of graduate education from the awarding institution IPEP**.

12/2023
- **Excellence Award by the Department of Physics, University of Patras**.
  Awarded for graduating top of the class with a grade of 9.35/10; selected to deliver the class oath as valedictorian.

04/2023
- **Scholarships for Outstanding Academic Performance by the Greek State Scholarships Foundation (IKY)**.
  Awarded for achieving the highest rank in my University for the academic year 2022-2023.

2021 –2022
- **Scholarships for Outstanding Academic Performance by the Greek State Scholarships Foundation (IKY) for students of vulnerable social groups**.

## Hobbies

- Painting, Knitting, Pottery, Hiking, Photography, Astronomy, Reading

## Volunteering

2022, *Patras*
- Panhellenic Astronomy Conference 2022

2021, *Patras*
- Non-profit Art In Progress, bringing contemporary Greek and international art to Patras.